HACKNOVA'26

PROBLEM STATEMENTS

PS-1

1. Title

EnergyIQ — Smart Energy Consumption Optimizer

2. Background / Context

Buildings, campuses, and commercial facilities consume large amounts of energy across different times of the day. However, raw energy-consumption data alone does not always reveal where energy is being wasted or when excessive consumption occurs.

An intelligent analytics system can transform smart-meter data into actionable insights by identifying unusual consumption patterns, peak-demand periods, and potential opportunities for reducing unnecessary energy usage.

3. Problem Statement

Develop a data-driven system that analyzes energy consumption data to identify **inefficient usage patterns, anomalies, and peak-load periods**, and provides actionable recommendations for improving energy efficiency.

The system should help facility managers understand **when, where, and why energy consumption is unusually high** and support better energy-management decisions.

4. Objectives / Goals

- Analyze historical and real-time energy-consumption patterns.
- Identify unusual or inefficient consumption behavior.
- Detect peak-load periods and high-demand intervals.
- Generate data-backed recommendations for reducing unnecessary energy usage.
- Present energy insights through an intuitive and easy-to-understand interface.

5. Target Users / Stakeholders

- Facility managers
- Building administrators
- Educational institutions and campuses
- Commercial and office buildings
- Energy-management teams

6. Constraints & Technical Considerations

- Recommendations must be supported by observed patterns in the provided data.
- The solution should be able to handle time-series energy-consumption data.

- Anomaly detection should distinguish unusual consumption from normal variations where possible.
- Visualizations should clearly communicate consumption trends, peaks, and anomalies.
- The solution should remain feasible to develop and demonstrate within the hackathon duration.

7. Resources Provided

- Synthetic smart-meter energy-consumption dataset with relevant fields and descriptions.
- **UCI Appliances Energy Prediction Dataset:** Time-series appliance energy consumption, temperature, humidity, and environmental measurements.
  Dataset: https://archive.ics.uci.edu/dataset/374/appliances+energy+prediction
- **UCI Individual Household Electric Power Consumption Dataset:** Household electricity measurements, including active power, voltage, current intensity, and sub-metering values.
  Dataset:https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption
- **UCI Power Consumption of Tetouan City Dataset:** Electricity consumption measurements from three distribution networks, suitable for consumption trend analysis and peak-demand identification.
  Dataset: https://archive.ics.uci.edu/dataset/849/power+consumption+of+tetouan+city

Teams may use their own datasets, publicly available datasets, or other relevant data sources, provided they are suitable for the problem statement and comply with applicable access and licensing requirements.

8. Expected Solution / Key Features

The solution is expected to include:

- **Consumption Analytics:** Analyze energy usage across time periods and identify consumption trends.
- **Peak-Load Analysis:** Identify periods of unusually high or peak energy demand.
- **Anomaly Detection:** Detect abnormal consumption patterns compared with expected usage.
- **Trend Analysis:** Visualize changes in consumption over time and identify recurring patterns.
- **Energy Insights:** Highlight major consumption drivers and inefficient usage patterns.
- **Recommendation Engine:** Generate practical recommendations based on the detected patterns.
- **Interactive Dashboard:** Present key metrics, trends, anomalies, peak periods, and recommendations in a clear visual format.

9. Expected Deliverables

Teams must submit:

- A **functional EnergyIQ dashboard** for exploring energy-consumption data.
- An **analytics engine** implementing consumption analysis, anomaly detection, and peak-load analysis.
- A **recommendation component** that generates data-supported energy-efficiency suggestions.
- A brief **technical explanation** of the approach, analysis methodology, and key insights obtained from the dataset.

10. Evaluation Criteria

- Quality and relevance of energy-consumption insights.
- Accuracy of consumption analysis, anomaly detection, and peak-load identification.
- Clarity and effectiveness of visualizations.
- Practicality and data-grounding of energy-efficiency recommendations.
- Overall functionality and usability of the solution.

PS-2



1. Title

Disaster Information Verification Agent

2. Background / Context

During disasters and emergency situations, information spreads rapidly through social media, messaging platforms, news sources, and public alerts. However, this information may be outdated, conflicting, incomplete, or false, making it difficult for responders and authorities to determine what information can be trusted.

An intelligent verification system can help analyze incoming claims, compare them against trusted sources, and provide evidence-based assessments to support faster and more informed decision-making.

3. Problem Statement

Build an **AI-powered information verification agent** that analyzes disaster-related claims and compares them against trusted emergency alerts, geographic information, weather data, timestamps, and other available reports.

The system should identify potentially unreliable or conflicting information, assess the freshness and credibility of available evidence, and present the supporting evidence clearly to emergency stakeholders.

4. Objectives / Goals

- Detect potentially questionable or unreliable disaster-related claims.
- Compare information across multiple available sources.
- Track the freshness and timestamp of information.
- Identify contradictions or inconsistencies between reports.
- Provide evidence supporting the verification result.
- Clearly communicate uncertainty when sufficient evidence is unavailable.
- Assist emergency stakeholders in making better-informed decisions.

5. Target Users / Stakeholders

- Emergency responders
- Government and disaster-management authorities
- Hospitals and healthcare organizations
- NGOs and relief organizations
- Citizens and community coordinators

6. Constraints & Technical Considerations

- The system must clearly communicate uncertainty when evidence is insufficient.
- The agent must not make unsupported factual claims.
- Verification results should be grounded in the available sources.
- Source timestamps should be considered when determining information freshness.
- Geographic information should be used where relevant to validate claims.
- The system should distinguish between verified, conflicting, outdated, and unverified information.

7. Resources Provided

- Synthetic social-media posts containing disaster-related claims, including potentially false, outdated, or conflicting information.
- Synthetic emergency alerts, structured incident reports, timestamps, and source metadata for testing the verification workflow.
- **CrisisMMD Dataset:** Disaster-related social media posts and images with humanitarian information annotations.
  Link: https://crisisnlp.qcri.org/crisismmd
- **GDACS (Global Disaster Alert and Coordination System):** Global disaster alerts and event information for cross-checking reported incidents.
  Link: https://www.gdacs.org/
- **ReliefWeb:** Humanitarian reports, disaster updates, and situation reports for gathering supporting evidence.
  Link: https://reliefweb.int/
- **NDMA India (National Disaster Management Authority):** Official Indian disaster-management information, guidance, and alerts.
  Link: https://ndma.gov.in/
- Teams may use their own datasets, publicly available datasets, official alerts, news reports, and other relevant sources, provided they are suitable for the problem statement and comply with applicable access and licensing requirements.

8. Expected Solution / Key Features

The solution is expected to include:

- **Claim Extraction:** Identify key factual claims from incoming disaster-related information.
- **Source Comparison:** Compare claims against available emergency alerts, reports, weather data, and other trusted sources.
- **Information Freshness:** Analyze timestamps and identify outdated information.
- **Contradiction Detection:** Identify conflicting claims or inconsistencies between sources.
- **Geographic Verification:** Map claims to relevant locations and identify geographic inconsistencies.
- **Evidence-Based Verification:** Provide supporting and contradicting evidence for each claim.
- **Verification Status:** Categorize information as verified, conflicting, outdated, or unverified based on available evidence.
- **Evidence Dashboard:** Present claims, verification status, sources, timestamps, and supporting evidence in an easy-to-understand interface.
- **Interactive Map:** Display geographically relevant claims, alerts, and verified information.

9. Expected Deliverables

Teams must submit:

- A **functional Disaster Information Verification Agent**.
- An **emergency information dashboard** displaying verification results and supporting evidence.
- An **interactive map interface** showing geographically relevant information.
- A **claim verification and evidence-analysis component**.
- A brief **technical explanation** describing the agent workflow, verification methodology, and handling of conflicting or uncertain information.

10. Evaluation Criteria

- Accuracy and quality of information verification.
- Reliability and grounding of sources and supporting evidence.
- Ability to handle information freshness and conflicting reports.
- Explainability and clarity of verification results.
- Usability and effectiveness of the dashboard and map interface.

PS-3



1. Title

SquadSync — Spotting Workload Imbalance and Integration Risk from Git and Task Data

2. Background / Context

Student project teams often discover collaboration problems only when deadlines are approaching. Work may be distributed unevenly, code reviews may be missed, or team members may delay integrating their work until the final stages of a project.

Raw commit counts alone do not provide a reliable picture of contribution or collaboration. A more meaningful analysis should consider factors such as task distribution, review participation, code activity, and integration patterns.

3. Problem Statement

Build a system that analyzes **Git repository activity and task-board data** to identify collaboration risks such as workload imbalance, insufficient review participation, and delayed integration.

The system should go beyond raw commit counts by using meaningful indicators and presenting clear evidence that helps student teams and mentors understand potential collaboration issues and take corrective action.

4. Objectives / Goals

- Identify meaningful collaboration patterns from Git and task-board activity.
- Detect potential workload imbalance among team members.
- Identify missing or uneven review participation.
- Detect delayed or risky integration patterns.
- Explain the evidence behind each identified risk.
- Use multiple meaningful indicators rather than relying solely on commit counts.
- Present actionable insights that teams and mentors can understand easily.

5. Target Users / Stakeholders

- Student project teams
- Teaching assistants
- Project mentors
- Faculty members supervising group projects

6. Constraints & Technical Considerations

- Use the provided synthetic or anonymised Git and task-board data.
- Contribution should not be assessed using raw commit count alone.
- Selected metrics and indicators must be clearly justified.
- The system should distinguish between genuine collaboration risks and normal differences in team activity.
- Only information necessary for team-level insights should be displayed.
- The solution should be feasible to implement and demonstrate within the hackathon duration.

7. Resources Provided

- **GitHub REST API:** Access repository commits, pull requests, code reviews, branches, and merge activity.
  Link: https://docs.github.com/en/rest
- **GitHub Archive:** Historical public GitHub event data that can support the analysis of development activity and collaboration patterns.
  Link: https://www.gharchive.org/
- **Apache DevLake:** An open-source engineering analytics platform for collecting and analyzing development activity from supported tools, including GitHub and Jira.
  Link: https://devlake.apache.org/

Teams may use their own Git repository data, task-board exports, publicly available datasets, or other relevant data sources, provided they are suitable for the problem statement and comply with applicable access and licensing requirements.

8. Expected Solution / Key Features

The solution is expected to include:

- **Data Ingestion & Normalisation:** Process and combine Git activity and task-board data into a consistent format.
- **Workload Analysis:** Analyze task distribution and development activity across team members.
- **Review Participation Analysis:** Identify patterns in code-review participation and potential review gaps.
- **Integration Risk Analysis:** Detect delayed merges, long-lived changes, or patterns indicating potential integration problems.
- **Collaboration Indicators:** Generate meaningful metrics that provide a broader view of team collaboration beyond commit counts.
- **Risk Detection:** Highlight potential collaboration risks based on the selected indicators.
- **Evidence-Based Insights:** Provide short explanations showing why a particular risk was identified.
- **Collaboration Dashboard:** Present workload distribution, review participation, integration status, and identified risks in a clear interface.
- **Mentor/Team View:** Provide actionable insights that can help teams and mentors address identified issues.

9. Expected Deliverables

Teams must submit:

- A **functional SquadSync collaboration-monitor prototype**.
- A **dashboard or report** displaying collaboration indicators and identified risks.
- A **data-analysis component** for Git and task-board activity.
- **Source code and documentation**.
- A **demonstration** showing how the system identifies collaboration risks.

- A brief **justification of the selected metrics and indicators**.

10. Evaluation Criteria:

- Meaningfulness and justification of the selected collaboration indicators.
- Ability to identify realistic workload, review, and integration risks.
- Ability to provide clear evidence behind identified risks.
- Usability and usefulness for students and mentors.
- Engineering quality and clarity of presentation.

PS-4



1. Title

Meeting-to-Execution Automation Agent

2. Background / Context

Meetings often result in important decisions, commitments, and follow-up actions. However, these outcomes are not always converted into clearly defined tasks with assigned owners and deadlines. Action items may be scattered across meeting notes, chat messages, and project management tools, making them difficult to track and manage.

An intelligent automation system can bridge the gap between discussions and execution by extracting actionable information from meeting transcripts or notes, organizing it into structured tasks, and routing it through an approval workflow before execution.

3. Problem Statement

Develop an AI-powered system that converts meeting transcripts or notes into structured decisions, action items, assigned owners, deadlines, and follow-ups. The system should distinguish confirmed commitments from suggestions and unresolved questions, identify duplicate or conflicting tasks, and support an approval workflow before assigning tasks or sending notifications.

The solution should help teams transform meeting outcomes into actionable, trackable tasks while reducing manual effort and preventing important commitments from being overlooked.

4. Objectives / Goals

- Extract decisions, action items, owners, and deadlines from meeting notes or transcripts.
- Distinguish confirmed commitments from suggestions and unresolved questions.
- Identify duplicate or conflicting tasks to improve task accuracy.
- Enable users to review and approve proposed tasks before execution.
- Integrate with a project management tool or export tasks in a structured format.
- Track task completion and highlight overdue or unassigned actions.
- Reduce the gap between meeting discussions and actual task execution.

5. Target Users / Stakeholders

- Project managers and team leaders
- Software development teams
- Corporate teams and business professionals
- Meeting coordinators and administrative teams
- Organizations managing multiple projects and follow-up activities

6. Constraints & Technical Considerations

- The system should support meeting transcripts or notes as input.
- Extracted tasks, owners, and deadlines should remain consistent with the provided meeting content.
- Confirmed commitments must be distinguished from suggestions and unresolved questions.
- Ambiguous task assignments should require user confirmation before execution.
- Notifications and task assignments should not be triggered without the required approval.
- Duplicate and conflicting tasks should be identified where possible.
- Integration with a project management tool or structured task export should be supported.
- The solution should remain feasible to implement and demonstrate within the hackathon duration.

7. Resources Provided

- **AMI Meeting Corpus:** A research dataset containing meeting recordings, transcripts, and annotations for studying meeting understanding and action-item extraction.
  Link: https://groups.inf.ed.ac.uk/ami/corpus/
- **QMSum Dataset:** A meeting summarization dataset containing meeting transcripts and query-based summaries, useful for testing meeting understanding and information extraction.
  Link: https://github.com/Yale-LILY/QMSum
- **Trello REST API:** Enables integration with Trello boards to create tasks, assign members, and manage due dates, subject to API access and permissions.
  Link: https://developer.atlassian.com/cloud/trello/rest/
- **GitHub REST API:** Supports task and issue management through GitHub repositories, including issue creation, assignment, and tracking.
  Link: https://docs.github.com/en/rest/issues

Teams may use their own meeting transcripts, notes, datasets, project management tools, or other relevant resources, provided they are suitable for the problem statement and comply with applicable access and licensing requirements.

8. Expected Solution / Key Features

The solution is expected to include:

- Meeting Input: Allow users to upload or provide meeting transcripts and notes.
- Decision and Task Extraction: Identify decisions, action items, owners, deadlines, and follow-ups.
- Commitment Classification: Separate confirmed commitments from suggestions and unresolved questions.
- Duplicate and Conflict Detection: Identify potentially duplicated tasks or conflicting commitments.
- Approval Workflow: Allow users to review, edit, approve, or reject proposed tasks before execution.
- Task Integration and Export: Send approved tasks to a project management tool or export them in a structured format.
- Confirmation Mechanism: Request clarification when task ownership or other important details are ambiguous.
- Task Tracking: Monitor task completion and highlight overdue or unassigned actions.
- Follow-up Management: Surface pending actions and follow-ups to improve accountability.

9. Expected Deliverables

Teams must submit:

- A functional application that converts meeting transcripts or notes into structured tasks.
- A task extraction component identifying decisions, action items, owners, deadlines, and follow-ups.

- A classification mechanism distinguishing commitments, suggestions, and unresolved questions.
- A duplicate and conflict detection component.
- An approval workflow for reviewing and confirming proposed tasks.
- A project management integration or structured task export mechanism.
- A demonstration showing transcript upload, task generation, approval, and transfer to a project board with owners and due dates.
- A brief technical explanation of the approach, workflow, and key implementation decisions.

10. Evaluation Criteria

- Accuracy of information extraction.
- Correct classification of commitments and suggestions.
- Effectiveness of duplicate and conflict detection.
- Reliability of approval and confirmation workflows.
- Effectiveness of task tracking and integration.
- Overall functionality and usability.

PS-5



1. Title

Competitor Intelligence and Market Monitoring Agent

2. Background / Context

Businesses need to continuously monitor competitors to understand changes in product pricing, features, product launches, and market positioning. However, manually collecting information from multiple sources makes it difficult to identify significant developments, maintain historical records, and compare competitors over time.

An intelligent monitoring system can automate the analysis of permitted public information, detect meaningful changes, maintain a historical record, and generate evidence-backed reports. Such a system can help businesses stay informed about competitor activities and make better-informed strategic decisions.

3. Problem Statement

Develop a competitor intelligence platform that monitors permitted public sources, detects meaningful changes in competitor pricing, product features, product announcements, and market positioning, and maintains a dated history of observed changes.

The system should enable users to compare competitors against configurable product criteria, receive notifications about significant developments, and generate concise reports in which every reported claim is linked to its source and observation date.

4. Objectives / Goals

- Monitor selected public competitor webpages and permitted data sources.
- Detect meaningful changes in pricing, product features, product announcements, and market positioning.
- Maintain a dated historical record of observed changes.
- Compare competitors using configurable product criteria.
- Link reported claims to their original sources and observation dates.
- Notify users about significant competitor developments.
- Generate concise, evidence-backed competitor comparison reports.
- Support informed market research and competitive decision-making.

5. Target Users / Stakeholders

- Business owners and startup founders
- Product managers and product development teams
- Market research and business intelligence analysts
- Marketing and sales teams
- Competitive intelligence and strategy teams

- E-commerce businesses and pricing analysts

6. Constraints & Technical Considerations

- Data collection must use permitted public sources and comply with applicable website terms and access restrictions.
- The system should distinguish meaningful changes from minor or irrelevant webpage modifications.
- Observed information should be compared against previously recorded data to identify changes accurately.
- Historical records should include observation dates and relevant source information.
- Every reported claim should be traceable to a verifiable source.
- Competitor comparisons should follow configurable product criteria.
- Notifications should prioritize relevant changes and minimize unnecessary alerts.
- The system should handle missing, inconsistent, or changing webpage content where possible.
- The solution should remain feasible to develop and demonstrate within the hackathon duration.

7. Resources Provided

- **Internet Archive Wayback Machine:** https://web.archive.org/ — Historical webpage snapshots, where available, for examining changes over time.
- **Common Crawl:** https://commoncrawl.org/ — Public web crawl data for research and analysis.
- **Kaggle Datasets:** https://www.kaggle.com/datasets — Potential product and pricing datasets for prototyping, subject to dataset availability and licensing.

These resources are optional. Teams may use suitable public sources, create sample data to demonstrate change detection, or use external datasets, APIs, and tools, subject to access and licensing requirements.

8. Expected Solution / Key Features

The solution is expected to include:

- **Competitor Monitoring:** Monitor selected competitor webpages and permitted public data sources.
- **Change Detection:** Identify changes in pricing, product features, announcements, and market positioning.
- **Historical Tracking:** Maintain dated records of observed information and detected changes.
- **Competitor Comparison:** Compare competitors against configurable product attributes and evaluation criteria.
- **Evidence-Linked Reporting:** Link reported claims to their sources and observation dates.
- **Change Prioritization:** Identify significant developments and distinguish them from minor or irrelevant changes.
- **Notifications and Alerts:** Notify users about meaningful competitor updates.
- **Automated Report Generation:** Generate concise reports summarizing competitor changes and comparisons.
- **Interactive Dashboard:** Present monitored competitors, historical changes, comparisons, and recent alerts in an accessible interface.

9. Expected Deliverables

Teams must submit:

- A functional competitor intelligence and market monitoring platform.
- A monitoring component for selected permitted public sources.

- A change-detection component for identifying relevant competitor updates.
- A historical record of observed information and detected changes.
- A competitor comparison component based on configurable product criteria.
- An evidence-linked reporting component with source links and observation dates.
- A notification mechanism for significant changes.
- An interface or dashboard for exploring competitor information and reports.
- A demonstration in which selected competitor pages are monitored, a simulated pricing or feature change is introduced, and a sourced comparison report showing the change is generated.
- A brief technical explanation of the approach, data sources, change-detection methodology, and key implementation decisions.

10. Evaluation Criteria

- Accuracy of data collection and monitoring.
- Effectiveness of change detection.
- Reliability of historical tracking.
- Quality of competitor comparisons.
- Accuracy of evidence and source links.
- Usefulness of alerts and reports.
- Overall functionality and usability.